Technical University of Denmark

Air Cleaner Insights: Optimize Filter Life-Time

Bachelor Project

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Author statement:

This project has been made in collaboration with the company Longhi Air[Int]. I extend my sincere appreciation to Longhi Air for their collaboration and permission to utilize their data, which have played a vital role in making the content of this report possible. Their contribution has been invaluable in ensuring the clarity and comprehensiveness of the information presented. It is important to point out that every image in this report that is not referred to or linked belongs to Longhi Air, an established organization that has provided me with the rights to use their visual content. All data used in this report is not publicly available. I also wanna thank my supervisor Bjarne Kjær Ersbøll for providing support with the report.

Abstract

Air quality has become a renown subject for its negative impact towards human health[Org]. To counter these impacts, companies such as Longhi Air have established air cleaner solutions to improve human health. Longhi Air is a Danish brand that develops and manufactures Air Cleaners with advanced technology and the highest quality.

However, cheap air cleaners are unable to measure power consumption. This inspired Longhi Air to discover ways to improve its air cleaners, such as identifying the precise time when renewing old filters within their machines. This report analyses data provided by the company to determine when the air filters have to be exchanged. This is done by looking at the effects that the filters can have on power consumption in the air cleaners. It is believed that the more contaminated the filter becomes, the more power consumption increases[Tea].

The analysis starts by comparing the trends in power consumption over a year-long active air cleaner in time-series graphs. This resulted in the opposite of the expected result. Measures to analyze anomalies are established using TimeGPT, a foundation model for analyzing time series data. Further research is needed to conclude the results.

Therefore, two experiments are performed to gain a deeper understanding of the results. The first experiment was done by deliberately making the filter dirty. This task involves bringing an air cleaner to a sanding room and comparing the before and after effects on power consumption. The latter experiment consists of simulating a clogged filter which is done by stopping the airflow in the air cleaner. This is done by putting a plastic bag over an air cleaner in a controlled environment.

The result of these experiments showcased that power consumption can be used to determine a need for change of the filter. On that note, it goes against our initial belief, where it is now discovered that it is with the decrease in power consumption that the need for the filter has to change. Although, of course, further research to determine this phenomenon will forever be needed. However, that is not the goal of this thesis.

A method is created to determine the need for a change in the filter. This is done with the help of curve-fitting methods and evaluation metrics are used to determine their effectiveness.

In conclusion, a method to indicate the precision of time to change filters is suggested and will be implemented by Longhi Air in the upcoming months.

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Acronyms

API Application Programming Interface. 8

CO_2 Carbon Dioxide. 34

ECO_2 Equivalent Carbon Dioxide. 34

GPU Graphics Processing Unit. 11

HEPA High Efficiency Particulate Air. 7

HTTP Hypertext Transfer Protocol. 8

LA Longhi Air. 5–8, 10, 16, 18, 20, 23–25, 27–31

MQTT Message Queuing Telemetry Transport. 8, 9

MSE Mean Squared Error. 13, 14, 29

NA Not Available. 18

NO_2 Nitrogen dioxide. 34

PCB Printed Circuit Board. 8

PM Particulate Matter. 34

PPB Parts Per Billion. 34

PPM Parts Per Million. 34

REST Representational State Transfer. 8

RPM Revolutions per minute. 6, 7, 16–19, 27, 30

RSS Residual Sum of Squares. 13

TVOC Total Volatile Organic Compounds. 34

1 Introduction

1.1 The Motivation behind the Project

Concerns over air quality have grown in recent years as a result of the negative impact of pollutants on human health and the environment[Org]. Companies such as Longhi Air (LA) are delivering new solutions to solve air quality challenges as the world deals with growing pollution levels. As early as 2001, LA has developed and sold smoking cabins to allow smokers to smoke in confounded places. Recently, LA has expanded their product lines to also include high-quality, high-capacity air cleaners/air purifiers. LA, has gained recognition for its advanced air cleaner technology aimed at creating healthier indoor environments[Int]. Moreover, this project is conducted in collaboration with the company LA.

LA's air cleaner technology allows for the collection of large amounts of data that are necessary for analyzing and improving air cleaning operations. Unlike cheap products, LA has the opportunity to measure power consumption and will therefore investigate ways to precisely calculate the right time for changing filters within its air cleaner systems. As of today, LA changes their filters based on a countdown timer. Through technical filter specification and real-life tests, LA has determined that a filter will be replaced after 8000 hours during air cleaners' active cycles. The lack of a precise method; for determining ideal filter replacement intervals can reduce the effectiveness and efficacy of these air cleaners. Especially when the switch is long-overdue. The accessibility to measure power consumption has launched an investigation aimed at finding a strategy for estimating and determining the ideal period for filter changes. This bachelor thesis examines data from LA's air cleaner technology to provide a solution to improve filter replacement.

Figure 1: Official logo of the Longhi Air company

1.2 Problem statement

The fan used by LA can monitor real-time power consumption, giving it a unique edge over cheaper air cleaners. Competitors and lower-cost solutions often receive a recommended time when the filter needs to be replaced, but it can vary depending on the circumstances of the filter. This frequently leads to untimely filter replacements, which either impose additional expenditures if replaced too early or even jeopardise the product efficacy when the change is overdue. The ability to assess power consumption in real-time gives LA an important edge over most competitors. This allows LA to base filter replacement timing on actual data. Hence, this improves the accuracy and cost-effectiveness of LA's air filtering systems vs rivals and is a competitive advantage.

This thesis will include an in-depth examination of the impacts that power consumption has on filters. The decision to investigate this aspect is based on available data collected from air cleaner monitoring and insightful discussions with LA stakeholders, which suggest the importance and relevance of examining power consumption as a primary analytical parameter.

The focus will be to investigate the power consumption changes of LA's air cleaners over time. One critical part of this investigation is determining whether there is a link between power consumption and the condition of the air filters. The purpose of this study is to discover if power consumption measurements can be used to determine when air cleaner filters have to be switched out.

1.2.1 Hypothesis

The hypothesis in this project is that power consumption can be used to determine when a filter needs to be changed. It is believed that as the filter becomes increasingly contaminated, so does the power consumption. The fan is set up to run at a constant speed using a built-in feature where the ventilator is controlled by a PI regulator[Y20] ensuring constant rotation speed(RPM). Therefore, it is assumed that the fan will require more power when the filter gets clogged because the resistance in the filter gets higher.

1.3 Report Structure

The following sections consist of a Method, Data, Data Analysis, Conclusion and a Discussion chapter. The purpose of the Method chapter acts as a guidance to ease the reader's comprehension regarding the thought process when performing this project. The data chapter describes the datasets and how the pre-processing of the different data occurs. The Data Analysis chapter delves into details of how the thought process of the analysis took place. The conclusion section summarizes our main findings and interpretations, whereas the discussion section discusses different aspects of the report.

2 Methods

The air cleaners of LA measure various factors. However, this analysis mainly focuses on power consumption over time, concentrating on the air cleaner's performance over five different operational levels. Each operational level has a unique RPM that requires varying power to achieve.

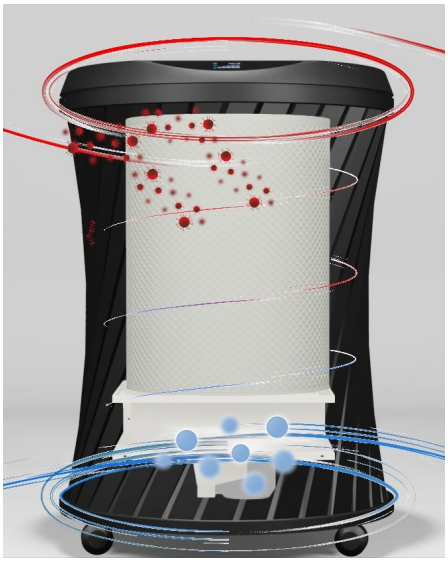
2.1 Background Knowledge

Before data can be analyzed, an understanding of LA's air cleaner's operations is relevant. This begins by considering how the air cleaners are built, followed by focusing on their controlling methods. Lastly, the how's and what's of the data are collected.

2.1.1 Air cleaner: Operational Insights

The air cleaner models can be seen in figure 2a, they are based on a Danish architectural design and built-in factories in Indonesia. The air cleaners draw air from all sides using a German 500W backwards curved centrifugal fan[ebm]. The air is then measured by an air quality sensor before entering the air cleaner ([Pla]). Following that, the air passes into the air cleaner through a pre-filter G3 ([Peand]), designed to catch larger particles. It then passes through the primary filter, namely a HEPA 13 filter ([Camnd]), which effectively catches the smaller particles. This will result the fan to disperse the filtered air. Clean air is released. The flow of the air particles can be seen in figure 2b where the colour red signifies dirty particles, while the colour blue signifies the opposite: clean particles.

(a) Types of Air Cleaners sold at LA



(b) Air particles and air filter seen in the air cleaner

Figure 2: Images of the air cleaner and airflow

2.1.2 Control and Connectivity of the Air Cleaner

The air cleaner's software is handled via an integrated touch screen that serves as the control interface. In addition, the customized touch screen is manufactured in China [Shend], and it is accompanied by a custom PCB built by LA.

The air cleaner has several control options. It may be controlled via the touchscreen interface, an app designed for mobile devices accessible via smartphones or tablets, or a web browser. The air cleaner has its own Wi-Fi access point and can also be connected to existing Wi-Fi networks.

Controllability of the air cleaner can be managed programmatically through a REST API (HTTP), enabling software to interact with and control the device. Furthermore, it can communicate with MQTT servers, for data collecting and providing additional control functions. An example configuration of the Wi-Fi setup can be seen in appendix 24.

It is noted that the fan assembly includes a motor and a sensor that measures power consumption as well as the speed level of the air cleaner. The connections and controls can be seen in figure 3.

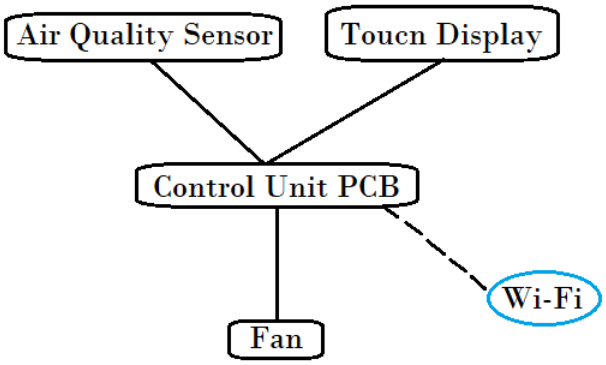
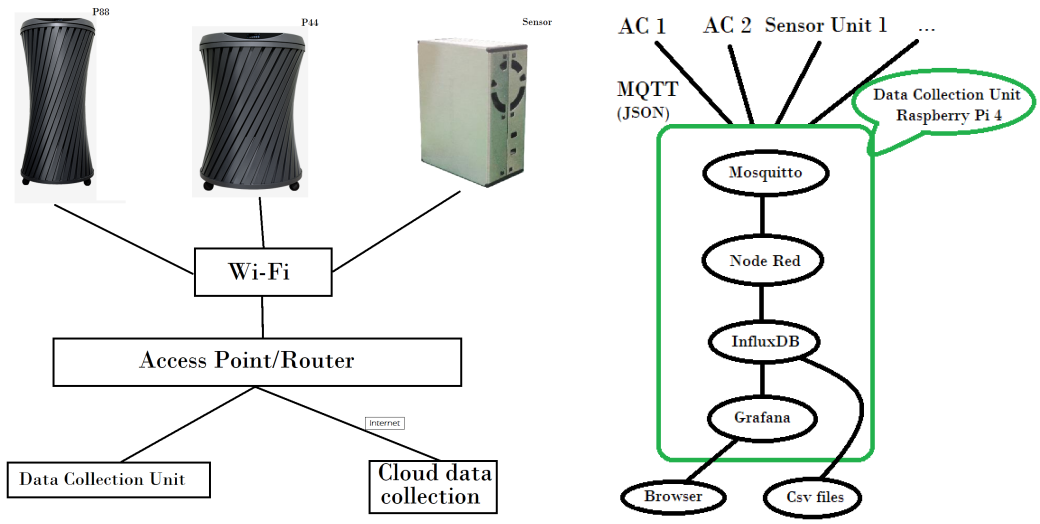


Figure 3: Overview of connections to control unit

2.1.3 Data Collection



(a) Overview of path to data collection

(b) Overview of path to data collection

Figure 4: Figures observer data collection

Figure 4a shows how air cleaners and sensor units can communicate with data collection systems. The air cleaners or stand-alone sensor units devices connect to a local Wi-Fi network, allowing communication through an access point or router. The connection enables the devices to send data to a central data-collecting unit, including air quality and power level data. Alternatively, the data might be transmitted via the internet to a cloud-based data gathering system for additional analysis and storage.

The sensor data is collected using the architecture shown in Figure 4b. The data collecting device(Raspberry Pi 4 [ZDNnd]), acts as a central system for storing and analyzing incoming data streams.

The data is initially sent through the Raspberry Pi 4 by wi-fi to an open-source MQTT server(Mosquitto [Lig17]). This server ensures that data streams are sent smoothly using the MQTT protocol. The MQTT server functions as a bridge, efficiently routing incoming data to succeeding components in the data-gathering process.

Following that, the data is processed using Node-RED [Nod], a visual programming tool built for IoT (Internet of Things) applications. Node-RED serves as a communication platform, allowing data to flow and be manipulated between various devices and systems. It allows for the construction of custom data processing procedures as well as the management of the data journey inside the data gathering unit.

After Node-RED, the processed data is saved in InfluxDB [Infed], an open-source database management system. InfluxDB specializes in time-series data management, providing for the effective storage and retrieval of historical data points.

Grafana [Graed], an effective open-source analytics and visualization tool, is used to display and analyze the data. Grafana uses data from InfluxDB to provide reports, data analysis, and dashboards. These dashboards provide employees with visually appealing representations and reports on many of the sensor values.

Furthermore, Grafana's functionality is available via a typical web browser, allowing users to easily see, interact with, and analyze data. Furthermore, data for this report was taken from InfluxDB(exported CSV file format), allowing for additional external analysis with Python.

2.2 Method Statement/Description

This section describes the methodical approach, to examine the idea that power consumption has an impact on determining when a filter needs replacement. The research begins with an analysis of real-life data gathered from an air cleaner located in a Jakarta, Indonesia, household that has been active for over a year. The raw data is pre-processed to extract relevant information and identify correlations. The goal of this step is to get an overview of the data and determine its usefulness while removing irrelevant data.

The air cleaner has five operational levels, each with a different power consumption average. Clogging of the filter will in an "average" environment not be measurable within a single day. Therefore we can easily aggregate data to the day level and this results in less but still fully valid data. The daily mean of power consumption is visualized in a graph over the time the air cleaner has been active. This visualization makes it easier to assess power consumption patterns over time. However, the initial analysis provides inconclusive results. To analyze this problem, TimeGPT[GM23], a transfer learning approach, is provided. TimeGPT is used to detect anomalies. Detecting anomalies helps make it more clear if there are any external factors influencing the measurements. TimeGPT does seem to find many anomalies and therefore factors such as temperature or humidity could influence the measurements.

As a result, an experimental setup is established at a facility owned by LA that manufactures air cleaners. During the experiment, a new air cleaner is purposely put in an extremely polluted area in a short time to clog the filter. The results are shown in a time-series manner once more, demonstrating a further contradiction to the expected results.

In response to these findings, a more controlled third experimental is being conducted at an office associated with LA. This experiment involves restricting the airflow of the filter with a carefully placed plastic bag. The results encourage an examination of the air filter system.

An in-depth investigation follows resulting in the development of curve-fitting formulas applicable to the five different operational levels. These formulas are a concrete solution that provides insights into optimizing power consumption together with air filter maintenance.

The method statement outlines the transition from real-world data analysis to controlled experiments, demonstrating the progressive nature of the scientific method used to investigate and in this case rejects the initial hypothesis, but based on the analysis we are still able to give a suggestion on how to solve the original problem.

2.3 Theory

After completing the methodical statement from real-life data analysis to controlled experiments, the following part looks into the theoretical foundations of two components used in this investigation: TimeGPT and curve fitting.

2.3.1 TimeGPT

It is important to note that TimeGPT used in this study is a closed beta version created by Nixtla [Nixnd]. Through a lucky connection, I was able to contact Nixtla, who invited me to be one of the first testers of their cutting-edge model. This special access enabled me to use TimeGPT's unique features to better analyze temporal trends and anomalies related to power consumption and air filter maintenance.

TimeGPT provides an exceptional solution for detecting anomalies because of its understanding of time complexities in sequential data. TimeGPT stands out from other approaches because it takes into account the chronological sequence and context of occurrences. Its unsupervised learning capability enables it to independently recognize abnormalities, making it especially useful in instances where deviations increase gradually over time.

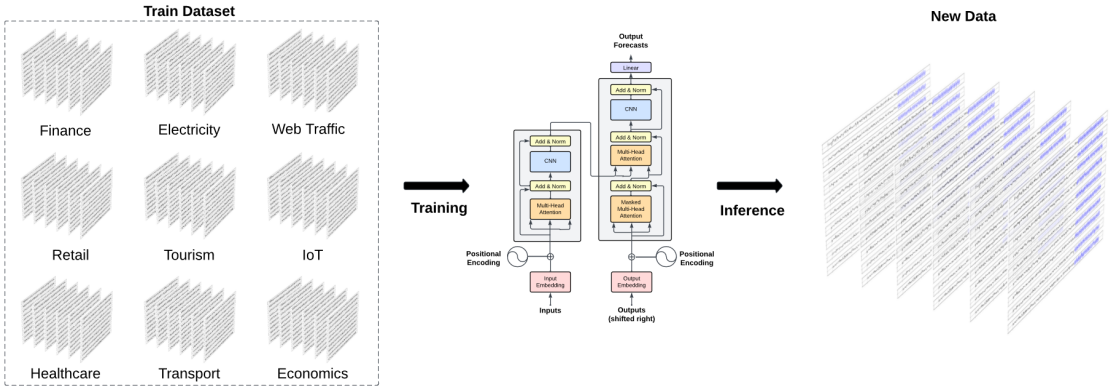


Figure 5: The structure of TimeGPT. Note: Taken from "TimeGPT-1" by Azul Garza, Max Mergenthaler-Canseco [GM23].

TimeGPT is offered as a foundation model that was trained on a diverse dataset of over 100 billion data points from several areas as seen in figure 5. The diversity of the training set allows TimeGPT to handle a variety of circumstances such as identifying patterns, correlations, and dependencies that go beyond specific tasks. The model's architecture allows it to easily adapt to different datasets and perform particular tasks as needed.

TimeGPT is a transformer-based model intended for time series data. It captures a wide range of temporal patterns through self-attention processes. The model's architecture enables it to properly manage the complex nature of time series data. It trained on a huge GPU cluster, maximizing hyperparameters like learning rates and batch sizes. The training set contained many time series with varying features.

Since TimeGPT is only used as a Transfer Learning Method for this report, only a brief explanation of the model is presented. For more information, a link to the official paper is provided [GM23].

2.3.2 Curve Fitting

Curve fitting is a mathematical technique used to approximate connections between variables in a collection of data points. This strategy enables us to model and evaluate empirical data, revealing underlying patterns and trends. In this part, there is described the theory of the essential principles and formulas connected with curve fitting such as linear, polynomial and exponential equations and their projection on the Least Square Method. There are also theories on the evaluation metrics and overfitting and underfitting that help determine the quality of the curve fit.

2.3.2.1 Curve Fitting Functions

Curve fitting functions, also known as regression functions, are mathematical models that estimate or show the connection between variables in a dataset. These functions seek to identify the best-fit curve that minimizes the difference between predicted values and actual data points. Curve fitting is the process of altering the parameters of functions to optimize the fit.

Linear Regression

Linear regression is a simple type of curve fitting that represents a linear connection between variables. The linear regression equation has this form:

$$y = mx + b \tag{1}$$

Where m is the slope and b is the y-intercept. Linear regression is useful for detecting basic patterns in which a straight line appropriately reflects the data[NWK85].

Polynomial Regression

Polynomial regression goes beyond linear regression to represent more complicated interactions. The general formula for polynomial regression with degree n is:

$$y = a_0 + a_1x + a_2x^2 + \ldots + a_nx^n \tag{2}$$

The fitting method determines the coefficients (a_0, a_1, $\ldots$, a_n). Polynomial regression can handle curves with many turning points [DS98].

Exponential Curve Fitting

Exponential functions are appropriate for applications in which the rate of change is proportional to the function's current value. An example of an exponential function's formula is:

$$y = ab^x \tag{3}$$

Parameter a represents the y-intercept or beginning value, which represents the function's starting point on the y-axis. Parameter b represents the growth or decay factor. Exponential curve fitting is useful for modelling processes that display exponential development or decline [MPV12].

I will in this report identify the best function to use to describe the available power consumption data.

2.3.2.2 The Least Squares Method

The Least Squares Method, a fundamental approach in regression analysis, is used to minimize the sum of squared residuals (RSS), indicating the differences between observed and expected values. As an example for exponential regression, the model is stated as $y = ab^x$, the primary goal is to determine the ideal parameters a and b that result in the minimum RSS. The mathematical definition of RSS is as follows:

$$\text{RSS} = \sum_{i=1}^{n} (y_i - ab^{x_i})^2 \tag{4}$$

Here, n is the number of data points, (x_i,y_i) are the observed pairings, and a and b are the parameters to be estimated. Minimizing RSS involves finding the optimal values for a and b, which is often accomplished by numerical optimization techniques.

The numerical optimization approach used in this thesis is the Levenberg-Marquardt algorithm[Mar63]. This iterative optimization approach is highly effective for nonlinear regression situations. The technique refines parameter estimates with each iteration, taking into account the model's local curvature in the parameter space. The parameter update equation is presented as follows:

$$\mathbf{p}_{\text{new}} = \mathbf{p}_{\text{old}} + \left(\mathbf{J}^T\mathbf{J} + \lambda\mathbf{I}\right)^{-1}\mathbf{J}^T\mathbf{r} \tag{5}$$

The updated and previous parameter vectors are denoted by $\mathbf{p}_{\text{new}}$ and $\mathbf{p}_{\text{old}}$, respectively. The Jacobian matrix is $\mathbf{J}$, the damping parameter is λ, the identity matrix is $\mathbf{I}$, and the residual vector is $\mathbf{r}$.

In the case of nonlinear regression situations, providing appropriate initial values is sometimes vital for the Levenberg-Marquardt process. These initial values serve as a starting point in the parameter space, affecting the algorithm's efficiency and precision in estimating model parameters while also helping to avoid entrapment in local minima.

2.3.2.3 Evaluation Metrics in Curve Fitting

When evaluating curve fitting quality, many measures are used to evaluate the agreement between observed and anticipated values. The three most widely used assessment measures are MSE and R^2.

Mean Squared Error (MSE)

The MSE measures the average squared difference between observed (y_i) and anticipated $(\hat{y}_i)$ values. For curve fitting, MSE computes the average of the squared residuals, giving an overall indicator of how well the model fits the data [Woh23].

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{6}$$

R-Squared (R^2)

R^2, also known as the coefficient of determination, measures how much of the variation in the dependent variable is predicted from the independent variable(s). It ranges between 0 and 1, with higher values suggesting a better match. R^2 is very important for determining the goodness of fit in regression models, which includes curve fitting [Woh23].

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{7}$$

2.3.2.4 Overfitting and Underfitting

After assessing the goodness of fit with metrics such as Covariance, MSE, and R^2, it is critical to address the possible problems of overfitting and underfitting in curve fitting [Gee23].

Overfitting

Overfitting happens when a model becomes too complex, collecting noise or anomalies unique to the training data but unable to generalize to new datasets. This typically leads to a high level of parameterization. Overfitting may be detected more easily by analyzing how well the model performs on unseen datasets. If the model performs well on the training set but struggles with additional data, it may be overfitted. Increased R-squared values on the training set, along with a substantial decline in performance on the validation set, may suggest overfitting. Additionally, a low but falling MSE on the training set followed by a growing MSE on incoming data might indicate overfitting.

Underfitting

Underfitting occurs when the model is extremely simple, failing to capture the underlying patterns in the data. This frequently happens when the model is too simple to accurately capture the underlying relationships. Underfit models typically perform badly on both training and fresh datasets. Low R-squared values and a high MSE on the training set indicate that the model is unable to explain the variation in the data. Similar low performance on a validation set demonstrates underfitting.

3 Data

The Python code used in this report can be found in the GitHub hyperlink: GitHub-Link

This section includes a look into available data, pre-processing measurements, and three data analysis methods used to evaluate the hypothesis.

3.1 Introduction to Data

In this subsection, there is a look at the data used for the analysis. Looking back at figure 4b, InfluxDB contains all available data and selected parts of that data can be converted into CSV files. Downloading this data from InfluxDB provides us with three different CSV files. Namely 'FanLevel', 'FanSpeedPower' and 'EnvSensor'. Two of these files are used in this report.

On that note, this project will consist of analyzing three different air cleaner models, namely AirMaster2000/M88, P88, and P44.

3.1.1 'FanLevel' dataset

The first CSV file used in this report is the 'FanLevel' dataset. This dataset contains important parameters to take into account before doing the analysis. To get a better understanding of this dataset, the CSV file is made into a data frame.

The 'FanLevel' dataset contains information on each transition at the operational level that occurs in the air cleaner. Each row provides information on the fan's activities, including a date, active status, identification code, fan level, fan name, and profile. Table 1 is an example sample from the 'FanLevel' dataset:

Table 1: Description of FanLevel

name	time	activeauto	id	level	name.1	profile
FanLevel	2022-07-29T03:43:27Z	0	6BE6E2E0	0	AirMaster2000	2
FanLevel	2022-07-29T03:43:36Z	0	6BE6E2E0	2	AirMaster2000	2
FanLevel	2022-07-29T03:43:40Z	0	6BE6E2E0	0	AirMaster2000	2
FanLevel	2022-07-29T04:00:31Z	0	6BE6E2E0	2	AirMaster2000	2
...	...	...	...	...	...	...
FanLevel	2023-08-28T13:15:45Z	1	6BE6E2E0	2	M88	1

The columns in table 1 represent the following information:

- **name**: The dataset name (i.e., "FanLevel").
- **time**: The timestamp when the data was recorded.
- **activeauto**: The fan's auto-activity status (0 for inactive, 1 for active).
- **id**: The unique identification code associated with the fan.
- **level**: The specific operational level of the fan. (0 - off, 1-5 levels)
- **name.1**: Name of the fan.
- **profile**: Profile of the fan(0 - Low, 1 - normal, 2 - High).

Data is sent from the air cleaner when a change in operational speed is made, as opposed to other datasets that are sent with a fixed sampling frequency.

In table 1, the 'activeauto' column is a function that lets the air cleaners decide their own operational level based on the pollution observed in their surroundings. In the 'name.1' column it can be seen that the name of the AirMaster2000 model switches to M88. This isn't an issue due to 'id' still being the same. This 'id' is unique per air cleaner while the name can be changed in the air cleaner. In the 'level' column, there are 6 different operational levels. Each level corresponds to a unique 'RPM' value.

This dataset gives insights into the operational behaviour, state of fans over time and profile performance.

3.1.1.1 Profile Level

An air cleaner can be configured to use one of three profiles: low, normal and high. This setting is used if an air cleaner is placed in a highly polluted area(profile level = high), or if an air cleaner is in an environment where the air cleaning capacity is not needed to be high(profile level = low). In low compound to high the 5 performance levels(RPM) will be offset to lower RPM and the opposite for high. A side effect is that the noise level on 'high' will be relatively higher than 'low' but have a higher cleaning capacity(Higher RPM for all 5 performance levels).

The table 1, contain a "profile" column categorizes the air cleaner's performance levels as low, normal, or high. Following an inspection of code from LA, the following values for each performance profile are seen in table 2.

Profile	**Level 0**	**Level 1**	**Level 2**	**Level 3**	**Level 4**	**Level 5**
Low	0	10,000	18,750	27,500	36,250	45,000
normal	0	15,000	23,750	32,500	41,250	50,000
High	0	20,000	30,000	40,000	50,000	65,534

Table 2: Levels for different profiles

If it is assumed that the max value 65,534 is 100% of the power that the air cleaner can utilize. Then the other values percentage can also be computed. This is seen in table 3.

Profile	**Level 0**	**Level 1**	**Level 2**	**Level 3**	**Level 4**	**Level 5**
Low	0.00%	15.28%	28.62%	42.00%	55.37%	68.75%
normal	0.00%	22.90%	36.28%	49.60%	62.92%	76.24%
High	0.00%	30.52%	45.76%	61.04%	76.24%	100.00%

Table 3: Percentage values for different levels and profiles

Table 3 shows the proportion of each value compared to the highest value in the dataset. It explains the relative size of each value with the maximum value observed across all of the performance profiles of the air cleaner. In this report, there is only worked with 'normal' performance. The other performances are removed from the data.

3.1.2 'FanSpeedPower' dataset

The 'FanSpeedPower' dataset contains measurements of fan power, speed set-points and actual speed over time. An example from the 'FanSpeedPower' dataset can be seen in table 4.

Table 4: Description of FanSpeedPower

name	time	id	name.1	power	$speed_{act}$	$speed_{set}$
FanSpeedPower	2022-07-29T04:01:41Z	6BE6E2E0	AirMaster2000	0	0	10
FanSpeedPower	2022-07-29T04:01:46Z	6BE6E2E0	AirMaster2000	14	600	616
FanSpeedPower	2022-07-29T04:01:52Z	6BE6E2E0	AirMaster2000	38	704	847
FanSpeedPower	2022-07-29T04:01:57Z	6BE6E2E0	AirMaster2000	44	974	975
...	...	...	...	...	...	...
FanSpeedPower	2023-09-02T08:47:05Z	1F988330	P88	46	975	975

The columns in table 4 represents the following information:

- **name**: The dataset name (i.e., "FanSpeedPower").
- **time**: The timestamp when the data was recorded.
- **id**: The unique identification code associated with the fan.
- **name.1**: Name of the fan.
- **power**: The power consumption of the fan in watts.
- **speed_act**: The actual speed of the fan in RPM.
- **speed_set**: The set speed for the fan in RPM.

Table 4 shows the power use patterns and speed shifts of various fans throughout time. This dataset will be analyzed to understand the link between power usage, fan speed, and their influence on total system performance. The 'id' column makes it possible to connect both datasets.

Another dataset named 'EnvSensor' is also available but isn't utilized in this report. It is described in Appendix 7.1.

3.1.3 Frequency of Data Measurement

The 'FanLevel' dataset records measurements anytime the fan level changes. This change occurs when an individual manually adjusts the fan level or when the system's automated function changes the fan level settings. As a result, the frequency of data entries in the 'FanLevel' dataset is determined by occurrences when fan levels change.

In the instance of the 'FanSpeedPower' dataset, data is collected at 5-second intervals.

The 'EnvSensor' dataset, on the other hand, collects environmental sensor readings at a lower frequency, with data observations happening every 10 seconds.

An overview of this is seen in table 5.

Table 5: Summary of Dataset Characteristics and Frequency of Measurements

Dataset Name	Main Characteristics	Frequency of Measurements
FanLevel	Captures fan level changes	Whenever the level is changed
FanSpeedPower	Records fan power consumption and speed	Every 5 seconds
EnvSensor	Records environmental sensor readings	Every 10 seconds

The reason for the high sample frequency is a test version of the air cleaner software used.

Understanding measurement frequency differences is important for evaluating each dataset's time-stamps which could become problematic. For example, combining the datasets 'FanLevel' and 'FanSpeedPower' would produce many NA values. This is due to 'FanLevel's timestamp not being consistent but changed at random. The datasets typically will not be observed during the same time, meaning it will require pre-processing to make said data comparable. This is done by adding data values to up-sample or down-sample to enable the comparison of values between the datasets. Later on in this chapter, a method will be described to avoid matching the datasets directly.

3.1.4 Missing Data

Wi-Fi networking is used in LA air cleaners to provide remote monitoring of sensor data. Wi-Fi connectivity problems are common in places and data loss might occur as a result of connection issues between air cleaners and monitoring devices. As a result, missing data is widespread in the data. As we are looking for trends over time, it does not matter if we are missing data in small quantities over time.

3.1.5 Filter Simplification

A simplification was done to reduce the amount of data to be analyzed: The air cleaner has 2 filters (pre-filter and main filter). The pre-filter is cheap and is changed more frequently than the main filter to optimally protect the expensive main HEPA filter. In the analysis in this report, there is no focus on what filters are getting clogged as it would only complicate the experiments. The findings are still valid as the power consumption will behave the same over time no matter how many filters are used.

3.2 Pre-Processing

In this section, there is a look at some preprocessing done to the data. Pre-processing helps ensure that data is suitable for analysis, visualization and modelling.

3.2.1 RPM Speed Indicate Fan Level

A way to avoid combining datasets 'FanLevel' and 'FanSpeedPower' is analyzed. This is achieved by analyzing constant RPM values that could be used to determine an operational level.

In table 4, the column'*speed_set*' in the 'FanSpeedPower' dataset has unique values such as 0, 616, 975, 1335, 1695, and 2054. These values are the RPM the air cleaner can be set to for the 'normal' profile. This has been observed by changing the operational levels and looking at the RPM in the local-host browser which can be seen in appendix 22.

The observation also taught us that the precise values correspond directly to distinct fan levels. For example, the value 0 represents the air cleaner inactive, whereas the values 616, 975, 1335, 1695, and 2054 represent progressively increasing fan levels, denoted as levels 1, 2, 3, 4, and 5, respectively. This straight mapping establishes a link between the numerical numbers in the '*speed_set*' column and their corresponding fan levels. The correlation between fan level and speed set values can be seen in table 6.

Fan Level	**Speed Set Value**
1	616
2	975
3	1335
4	1695
5	2054

Table 6: Correlation between Fan Levels, Speed Set Values

Due to this correlation, it is possible to avoid combining the datasets 'FanLevel' and 'FanSpeedPower' to determine the fan level of each data point.

This correlation has provided a solution to divide the data into each respective fan level. Creating 5 different sub-data-frames to analyze for each of the upcoming scenarios.

This concludes that there are 5 'FanSpeedPower' datasets with a different operational fan level.

3.2.2 Matching RPM of Speed Set Value with Actual Value

The 5 different sub-data-frames contain all data points for their respective operational levels. The data points are only taken based on the speed set value. This means that the datasets still include data points where the actual speed has not reached the value of the speed set value, due to the fact that it takes some seconds for the fan to reach the new set-point. These data points are removed to only focus on relevant where we know the system is in a stable mode.

When '*speed_act*' matches '*speed_set*'(divided into sub-data frames for each level), it guarantees the correct detection of power consumption at those specific occurrences.

The 5 different sub-data-frames now only contain useful data.

3.2.3 Use of pre-processed data

The pre-processed sub-data-frames will be used for all experiments in the coming data analysis. The data frames make it possible to test different analyses for each operational level, visualize data and fit data into models for interpretation.

4 Data Analysis

This section attempts to prove the hypothesis by analyzing data from 3 different experiments. The first experiment analyzes real-life data. The second experiment analyzes data where the filter of an air cleaner is purposely clogged. The third experiment restricts air from entering the filter. Lastly, a predictive mode for filter replacement is suggested.

4.1 Analysis of Real-Life Data

To determine if power can be used to indicate a need for change in the filter, real-life data is observed.

The analysis starts with data from an air cleaner model M88. This model has the largest available data belonging to LA. The data taken from this model ran from September 2022 to September 2023, which is exactly a year. The air cleaner was placed in a private house. Therefore the dataset represents real-life measurements. This dataset was the only dataset available when the project was started.

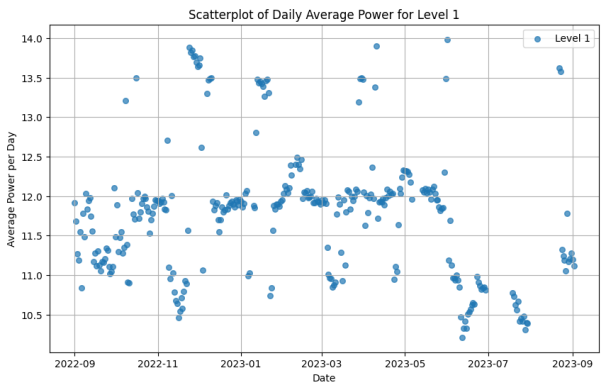
Starting with the 5 pre-processed 'FanSpeedPower' sub-data frames 3.2.2. The data contains 2604858 data points split by the 5 levels as seen in table 7.

Data Points	Count	Percentage of Count (%)
Level 1	2510983	96.40%
Level 2	59266	2.28%
Level 3	31482	1.21%
Level 4	2631	0.1%
Level 5	496	0.02%

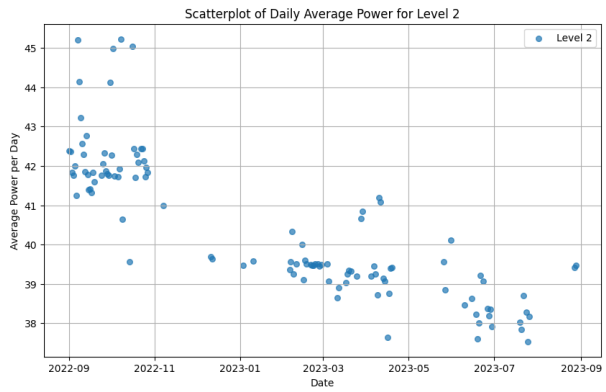
Table 7: Number of data points and percentage of count for different levels

In table 7, an observation is that the count decreases as the operational level increases. The air cleaner runs mostly 96.40% on level 1 and 2.28% on level 2. This is according to LA normal behaviour because of the air cleaner's large cleaning capacity.

The sub-data frames are aggregated into a daily mean of the power. By aggregating the power into daily mean, less data, but more relevant data can be observed. The reason it is more relevant is because we are looking for patterns in the difference on a day-to-day basis. The daily mean of power for each level over the 365 days is visualized in a scatterplot in figure 6,7 and 8.

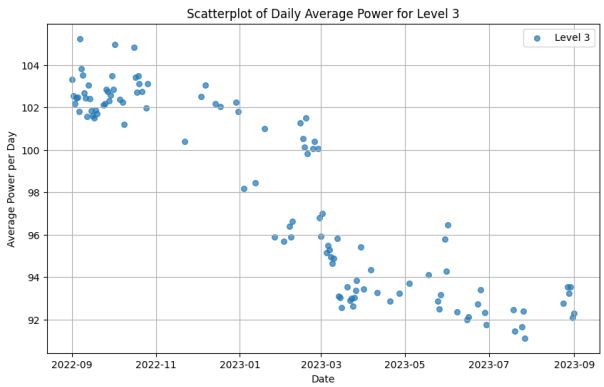


(a) Daily average mean of level 1

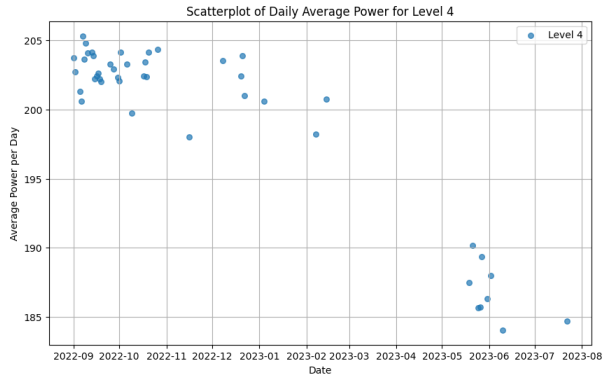


(b) Daily average mean of level 2

Figure 6: Snippet of the timestamp for level 1 & level 2



(a) Daily average mean of level 3



(b) Daily average mean of level 4

Figure 7: Snippet of the timestamp for level 3 & level 4

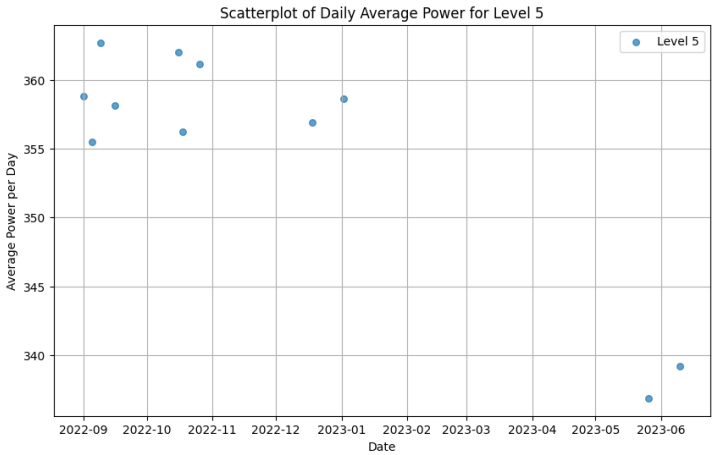


Figure 8: Daily average mean of level 5

From the figures 6,7 & 8, it can be seen that the data is fluctuating.

It can also be seen that the data becomes less and less reliable as the level increases due to the fewer amount of data points at those stages, where only a few times did the air cleaner go above level 2.

An unexpected finding in our investigation is that the higher the operational level, the more the power tends to decrease. This is not according to the hypothesis.

Determining anomalies might reveal an unexpected pattern or behaviour that could have lead to these unexpected results.

TimeGPT, a transfer learning approach designed exclusively for time-series data, is used to identify potential anomalies. These anomalies might prove that there are external factors that could cause power consumption to decrease. TimeGPT will exclusively be used on Level 1, due to the lack of data points in other levels. Through different tests, a confidence level of 90% is used to determine anomalies. The results are seen in figure 9.

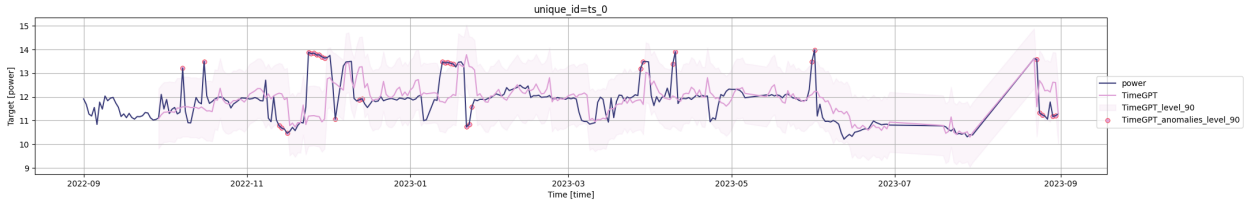


Figure 9: Anomalies in a confidence level of 90% for Level 1

From the 365 days in the dataset, 309 of them have the air cleaner been active with operational level 1. This means that there a 56 data points of missing values in this data frame. This could cause inaccuracy in detecting anomalies.

Figure 9, shows that there have appeared anomalies. Anomalies are when power deviates from its expected behaviour. These anomalies suggest that there is an external factor that has influenced the power. This could be factors such as temperature or humidity.

Because of these findings, it was decided to do another experiment. The filter might also still be performing ideally, so we cannot conclude anything with this dataset.

4.2 Experimental Setup: Air Cleaner Model P44 in Factory Environment (Dataset no.2)

To better understand the influence that contaminated filters have on power consumption, a new experiment was performed in a factory belonging to LA. The focus of this study was the air cleaner model P44 where the trial lasted two weeks. The experiment was done because no conclusions could be made based on the data analyzed in Chapter 4.1.

The first stage required creating a baseline for a new air cleaner. Afterwards, the air cleaner was exposed to a dusty particulate-rich environment. The air cleaner was set to level 5, the highest power setting, for 4 days to ensure a faster clog of the filter. This phase took place at the factory, where air cleaner was purposely exposed to the atmosphere found in an industrially busy and dusty environment.

Following the first dirt buildup period, power consumption data was collected for each level and compared to the baseline power consumption. This study sought to determine how increased filter dirtiness affected power utilization at various levels, so offering insight into the link between air filter condition and energy consumption.

Following the first examination, the air cleaner was put through additional testing in a sanding room to increase dust exposure. During this stage, an effort was made to stimulate an environment with an exceptionally high concentration of dust, allowing us to examine how the air cleaner responds to a more challenging filtering scenario.

The whole process of the experiment has been logged and can be seen in Appendix 15.

Figure 10 shows the power consumption data throughout the two-week experiment.

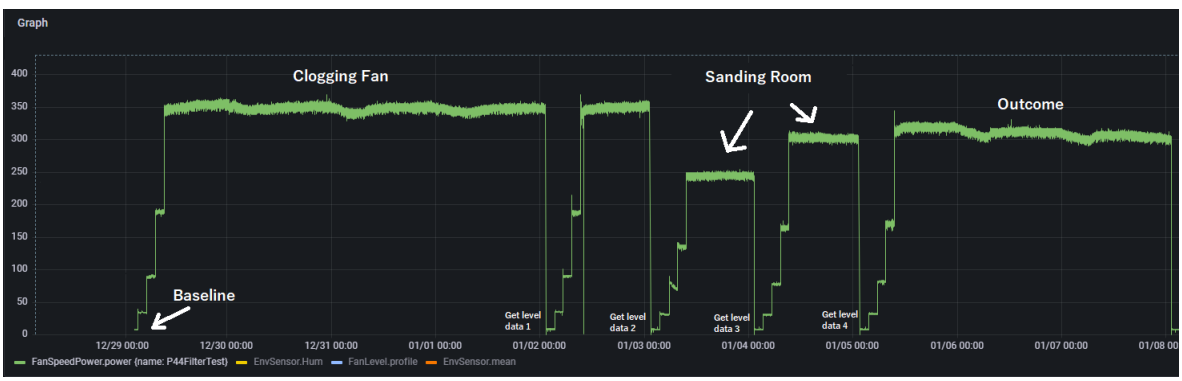


Figure 10: Graph of the timeline on the power consumption of the P44 air cleaner

In figure 10, during the clogging fan stage, it can be seen that the horizontal line is seen to shift up and down. We do not know what this is. It looks like it repeats every day. Maybe there is a correlation to e.g. temperature. This will not be investigated in this report, but it is recommended that LA should examine this further.

It is also seen in figure 10 that each horizontal in the graph corresponds to the levels of the air cleaner operations. The power consumption graph shows insignificant changes before exposure to the sanding chamber. This indicates an initial phase where the air cleaner's filter may not have gathered enough dust to become clogged. During the sanding room phase, there is a noticeable and constant drop in power usage across all levels. This unexpected reduction calls into question the initial belief, implying another link between dust exposure and power consumption decrease

that needs more examination. Relocating the air cleaner to its original position after sanding room exposure results in a significant reduction in power consumption.

It is seen that the power consumption is lower when the filter gets clogged. This was not expected according to the hypothesis in 1.2.1.

The air cleaner was placed in a sanding room for 2 days. The second day shows higher power consumption compared to the first day. This is because the filter was saturated early on the first day(lower power consumption) and on the second day in the morning, the filters were manually cleaned and only a little sanding was done that day. After the second day, the air cleaner was moved to a cleaner area and the filter was cleaned again and yet again power consumption increased.

The daily removal of the air cleaner's filter for photographic recording explains the increase in power consumption noticed throughout the day after the sanding stage. This action unintentionally removes collected dust, corresponding to a cleaner filter.

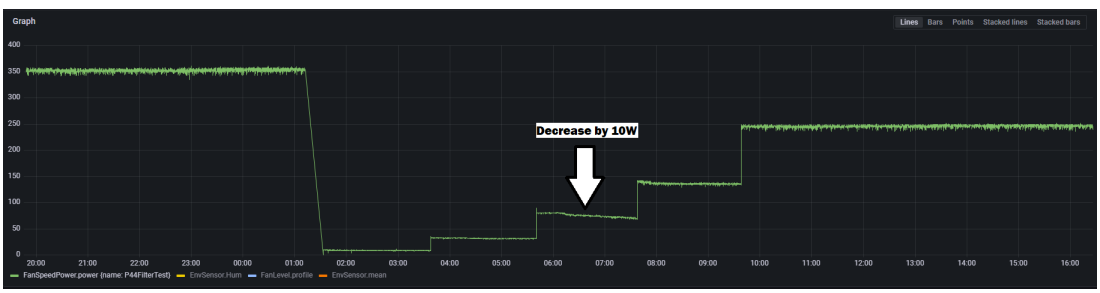


Figure 11: Graph of the timeline on the power consumption of the P44 air cleaner during first sanding room

In Figure 11 an anomaly develops at level 3 of the sanding room phase(first day), where power consumption suffers an unexpected drop of 10W. This is when the filter gets very clogged with particles.

This analysis has proved that power consumption can help determine the need for a filter change, but the initial belief was that the power consumption would increase. In the air cleaners of LA it seems that the power consumption decreases as the filter gets dirtier. Another more controlled experiment is performed to prove this statement.

4.3 Controlled Experiment in Longhi Air Office(Dataset no.3

The second experiment proved that there is a correlation between filter clogging and power. The initial belief was that power increases as the filter gets more clogged. This is not the case for LA's air cleaners. These air cleaners seem to have their power decrease as the filter gets more clogged. A 3rd experiment is now performed to test how much impact a clogged filter can have on power consumption.

This third experiment focuses on the effect of airflow limitation on power consumption simulating a clogged filter. For this experiment, the pre-filter was removed to ensure a focus on the main filter. A plastic bag was carefully positioned to obstruct airflow to the filter. This was performed in a controlled environment in an LA office to ensure no influence of external factors such as temperature and humidity.

The first stage was to set performance standards, the testing began with a baseline test on a new P88 air cleaner model, where the power was observed for each operational level. Each level was tested for around 5 minutes.

The following phase entailed gradually limiting airflow by obstructing portions of the 88cm air cleaner. The goal was to see how different degrees of airflow restriction affected the power consumption of the air cleaner.

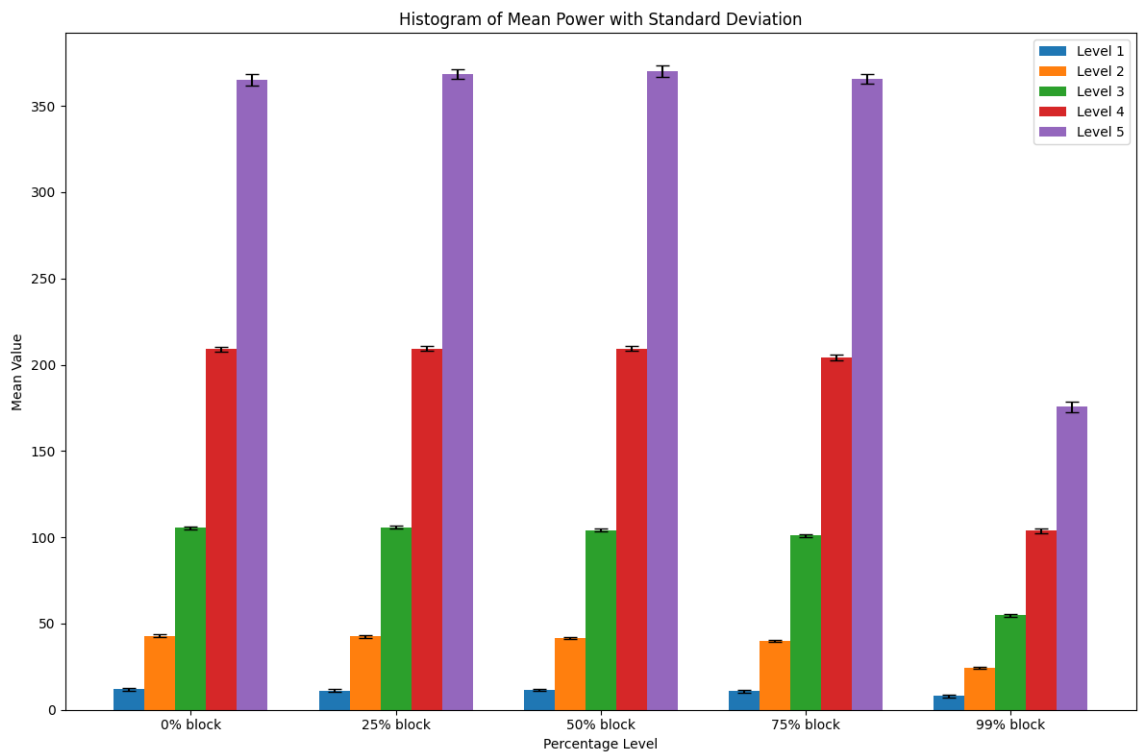


Figure 12: Histogram of Mean Levels including their Standard Deviation

Table 8: Mean and Standard Deviation for Each Level

	Level 1		Level 2		Level 3	
	Mean	Std Dev	Mean	Std Dev	Mean	Std Dev
0% block	11.88	0.82	42.97	0.76	105.43	0.68
25% block	11.21	0.78	42.54	0.78	105.74	0.78
50% block	11.40	0.66	41.68	0.71	104.23	0.84
75% block	10.80	0.81	39.89	0.69	101.06	0.83
99% block	8.00	0.78	24.48	0.70	54.71	0.89

	Level 4		Level 5	
	Mean	Std Dev	Mean	Std Dev
0% block	209.08	1.49	365.05	3.28
25% block	209.40	1.55	368.56	2.66
50% block	209.26	1.42	370.14	3.36
75% block	204.08	1.61	365.85	2.84
99% block	103.83	1.46	175.54	3.09

Figure 12 and table 8 summarize the mean power consumption and standard deviation values for each level of the air cleaner. The values ranging from '0% block' to '99% block,' represent varying degrees of airflow obstruction, providing insights into the air cleaner's performance at varied operational levels.

At Levels 1, 2 and 3, as airflow limitation increases, the mean power consumption decreases slightly. The standard deviation for these levels is also very low, indicating that all data is clustered very close around the mean. This means that data is very reliable.

At Levels 4 and 5 as airflow limitation increases, the mean drops more exponentially with a higher standard deviation. The standard deviation is still considered small.

Across all levels, the experiment consistently shows a drop in mean power consumption as airflow obstruction increases. It is seen that the higher the operational level, the more spread out the data becomes. Even though the standard deviations are still consistent around every level, adding to the dependability of the observed patterns.

To investigate this further, more tests on airflow obstruction were performed. This resulted in the visualization of figure 13.

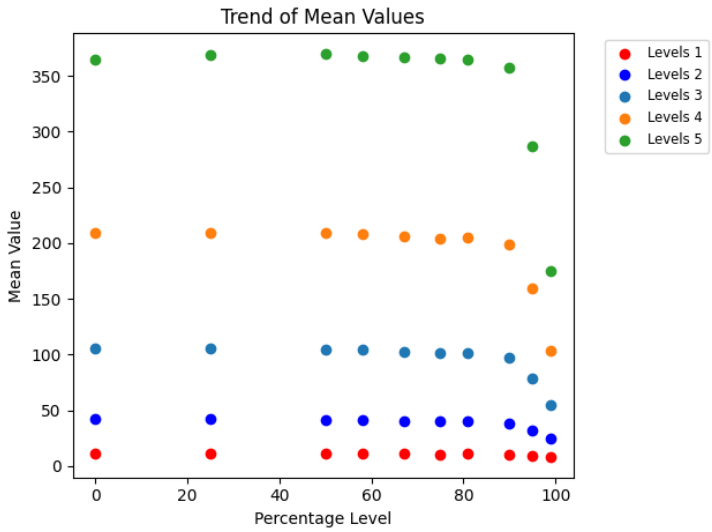


Figure 13: Trend of each mean value for each Level

Figure 13 shows 10 tests performed for each operational level. The x-axis indicates the percentage of the filter blocked, and the y-axis is the mean value of the power. It can be seen that at a certain clogging of the filter, around 80%, the power seems to drop exponentially. The higher the operational level, the more drastically the power drops.

This concludes that the hypothesis in this project has proven to be correct, but the initial belief that the power increases as the filter becomes more clogged is the opposite for the air cleaners of LA.

The reason for this requires further analysis. It is believed to have something to do with the motor of the fan. The fan is equipped with several mini processors and temperature sensors. Built-in algorithms are taking into consideration three internal motor temperatures, power reduction logic and other safety measures. As the laws of physics indicate we would need more power to maintain the same RPM with a clogged filter, there must be other factors probably related to the fan results in the measurements of lower power consumption. But as we have found a correlation between filter life-time and fan power consumption, we can still conclude we have a way of optimally indicating when a filter should be replaced. This is an opportunity that LA can decide to investigate.

4.4 Predictive Models for Filter Replacements

The study of the air cleaner data revealed a correlation between filter clogging and power consumption. This correlation can be taken advantage of to create a method to determine when the filter should be replaced in LA's air cleaners.

In this report, there is a proposed creation of prediction models with curve fitting for each operational level.

Figure 13 is used as a template for the suggestion of the most suitable functions. Looking at the 10 observed data points at each level, it can be seen that the data is almost linear until around 80% where the data points drop exponentially the higher the level.

That means the most suitable function to deal with in this case is the exponential function. Through testing functions with Desmos, a good exponential function for this model is:

$$a - b^{c \cdot x - d} \tag{8}$$

The parameter a is a constant that decides the y-intercept and in our case start of the function(power consumption of a clean filter). While b and c are the exponential decay factors. Lastly, d is included to shift the curve(percentage level=x-axis). Applying this function with curve-fitting to the values seen in figure 13 didn't work at first and that was due to the need for an initial guess of parameters(initial values). Initial guessing parameters help the process converge to a solution effectively. The parameters were decided with Desmos. Using curve.fit with the function(13) resulted in a function for each respective level as seen in figure 14.

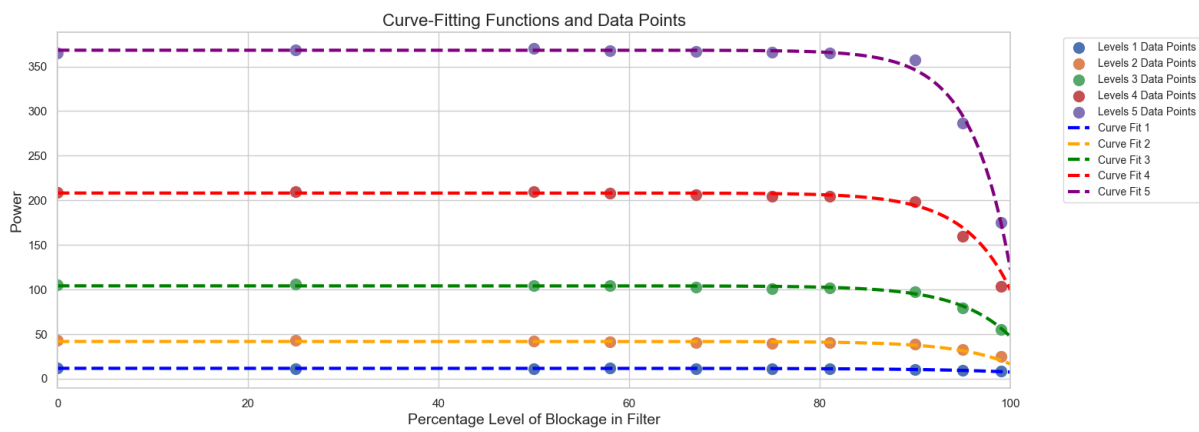


Figure 14: Curve-fit function on each mean value for each Level

The functions seen in figure 14, are shown in equation 9.

$$\text{Levels } 1 : 11.50 - 1.10^{(1.21 \cdot x - 106.02)} \tag{9a}$$

$$\text{Levels } 2 : 41.64 - 1.06^{(3.07 \cdot x - 251.51)} \tag{9b}$$

$$\text{Levels } 3 : 103.99 - 1.04^{(4.72 \cdot x - 368.88)} \tag{9c}$$

$$\text{Levels } 4 : 208.10 - 1.05^{(4.21 \cdot x - 324.91)} \tag{9d}$$

$$\text{Levels } 5 : 368.29 - 1.05^{(4.94 \cdot x - 381.15)} \tag{9e}$$

To see each function independently, look at appendix 7.3.

To make sure that these equations (9) can be useful to LA, evaluation metrics of each function are established.

Table 9: R-squared and MSE Values for Mean Levels

Operational Levels	**R-squared**	**MSE**
1	0.9537	0.0638
2	0.9763	0.7142
3	0.9907	2.2245
4	0.9932	7.2145
5	0.9941	20.9743

The table 9 provides insights into the reliability of the exponential curve fitting strategy in approximating the observed operational Levels. The R-squared values, which indicate the proportion of variability explained by the model, are consistently high throughout all Levels, ranging from 0.9537 for Levels 1 to 0.9941 for Levels 5. These figures show that the selected exponential function gives a good fit to the data, accounting for 95.37% to 99.41% of the variation in Levels. The growing trend in R-squared values with higher Levels indicates that the curve-fitting model is getting better at capturing the underlying patterns in the data. This is probably due to the fact that for lower levels(1 and 2) the variation in power from 0% to 100% is small compared to larger power variation on higher levels.

However, MSE values, which quantify the average squared differences between observed and projected values, show an increased trend with higher levels. While MSE is predicted to grow as the data becomes more complex, it is critical to evaluate these numbers carefully. The MSE values, which range from 0.0638 for level 1 to 20.9743 for level 5, show that the model's prediction accuracy decreases as Levels increase. As the level 5 power values are higher (0% power consumption is ? and only ? for level 1) than for level 1 the MSE is also expected to be higher at higher levels.

The high R-squared values across all levels show that the selected exponential function has great explanatory power, implying that the models successfully reflect the data's underlying patterns. These findings are very useful for LA since the correct prediction of levels allows for a meaningful assessment of air cleaner filter effectiveness. The models provide useful predictions that can help LA determine ideal filter replacement intervals.

This concludes that with the help of curve fitting, we have been able to create prediction models to determine filter replacement. The use of evaluation metrics has determined that the models are a good fit, and can therefore be used by LA.

The concluding functions can be seen in appendix 21.

5 Conclusion

This project is based on measurements from and experiments done at Longhi Air[Int], Jakarta, Indonesia. A fruitful collaboration was established with Longhi Air.

The project set out to determine if an optimal air cleaner filter change time can be determined based on changes in power consumption measurements from the fan over time.

Three datasets have been analyzed. The first dataset was from a residential house in Jakarta, Indonesia where data was collected from an air cleaner over a year. This dataset did not contain enough information which led to setting up an experiment in a polluted environment (factory) to try to saturate a filter in a short time while collecting measurements from the system. Data was collected and analyzed and a negative correlation between power consumption and clogging of the filter was found (a clogged filter requires less power to maintain a fixed speed(RPM) compared to a new unclogged filter). This was the opposite of what was thought to be true. The reason for this was not investigated because data indicated a solution to the hypothesis could still be found. To further investigate, a third experiment was set up in a controlled environment to eliminate the possible correlation between temperature and humidity. This experiment was performed in an air-conditioned office over a short time with various simulated clogged filters (by partly blocking a clean filter).

Analyzing the data from this 3rd experiment, curve-fitting was applied and a relationship (exponential decay) was found between the filter clogging level and the fan power consumption. Well-fitting models were found for each of the air cleaners' operating levels. It was shown that the problem can be solved by evaluating the fan power consumption over time.

With the findings in this report LA can now implement in their air cleaner control systems an algorithm to let the air cleaner owner know when the optimal time for filter change is due. This will ensure optimal cleaning of the air. Previously filter change time was determined by a timer which resulted in either the filter being changed too late (air cleaner will not work as expected) or too early (adding unnecessary cost to the customer).

LA is already working with additional tests and is modifying the air cleaner control software to use the findings from this report.

LA will investigate the finding where lower power consumption is measured instead of the expected higher power consumption for a clogged filter, but because of the findings in this report, LA will continue to use the same measurements and will consider adding a second power consumption reading to the data that can be collected from the air cleaners if a problem is found in the existing software or if there are issues with the internal fan calculations of power consumption.

6 Discussion

To fully understand why lower instead of increased power readings were seen with a clogged filter, additional experiments and investigations need to be done. Discussions with the fan manufacturer as well as further research on the multitude of parameters in the fan control system should be done. As mentioned in chapter 4.3, LA will look into this.

Looking at the data from the second experiment (placing an air cleaner in an extremely polluted room) some periodic patterns (day frequency) were observed. These observations should be analysed to obtain further knowledge of power consumption. There most likely is a correlation between power consumption and temperature and/or humidity. As we have access to temperature and humidity measurements in another dataset from the air cleaner (not used in this report) a correlation analysis could be made.

There are various ways to implement the findings from this report in the control software of the air cleaner. One method could be to let the air cleaner run through a sequence of changes in operating levels (1 to 5) when the air cleaner initially is set up and configured. Saving mean values of the power readings per operating level would provide a baseline for an algorithm based on the formulas found in this report. When a "clogging percentage" is defined by LA (for example if filters need to be changed when power consumption is 20% lower than the baseline readings) the air cleaner could test if this limit is reached and then indicate a filter change is needed. This can even be done with filter clogging percentages per operating level.

Another method is to create a dashboard in Grafana(The current panel used by stakeholders at LA) and include a detector meter that showcases the live percentage of the condition of the filter.

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7 Appendix

7.1 'EnvSensor' datasets

The 'EnvSensor' dataset contains a wide range of environmental sensor measurements collected over time.

An example of the 'EnvSensor' datasets can be seen in table 10.

Table 10: Description of EnvSensor

name	time	ECO2	Hum	$N0_3$	$N0_5$	$N10_0$	$N1_0$	$N2_5$	$N5_0$	NO2	$PM10_0$	$PM1_0$	$PM2_5$	TVOC	Temp	id	name.1
EnvSensor	2023-01-01T00:00:05Z	802	73.1	60	20	20	20	20	20	0	0	0	0	304	23.3	6BE6E2E0	AirMaster2000
EnvSensor	2023-01-01T00:00:07Z	721	66.3	729	211	10	54	20	20	0	7	3	6	223	30.2	A4E6E2E0	AP1Test
EnvSensor	2023-01-01T00:00:15Z	801	73.2	60	20	20	20	20	20	0	1	1	1	300	23.3	6BE6E2E0	AirMaster2000
EnvSensor	2023-01-01T00:00:17Z	711	66.3	741	216	10	53	20	20	0	7	3	6	215	30.2	A4E6E2E0	AP1Test
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
EnvSensor	2023-09-02T15:23:12Z	1246	45.6	264	88	20	31	20	20	0	2	0	2	2154	22.9	6BE6E2E0	M88

The columns in table 10 represents the following information:

- **name**: The dataset name (i.e., "EnvSensor").
- **time**: The timestamp when the environmental data was recorded.
- **ECO_2**: The level of CO_2 in PPM.
- **Hum**: The humidity percentage.
- **N0_3, N0_5, N10_0, N1_0, N2_5, N5_0**: The presence of specific particle sizes in the air.
- **NO2**: The level of NO_2 in the air.
- **PM10_0, PM1_0, PM2_5**: The concentration of PM of specific sizes in micro-grams per cubic meter.
- **TVOC**: TVOC in PPB.
- **Temp**: The temperature in degrees Celsius.
- **id**: The unique identification code associated with the sensor readings.
- **name.1**: The associated name of the model.

This dataset gives important insights into numerous environmental elements, such as air quality indices and temperature, which are critical for analyzing indoor environmental conditions and possible health and comfort consequences.

7.2 Observations from experiments

Air Cleaner Test

No	Location	Time (WIB) (Date)	Total Time	Speed (Normal)
1	Smoke Cabin Production	(28/12/2023) 15:00 ~ (29/12/2023) 08:30	17 hours	5
2	Air Filter Production	(29/12/2023) 08:30 ~ 10:30	96 hours	1
		10:30 ~ 12:30		2
		12:30 ~ 14:30		3
		14:30 ~ 16:30		4
		16:30 ~ 08:15(02/01/2024)		5
3	Air Filter Production	(02/01/2024) 08:15 ~ 10:15	24 hours	1
		10:15 ~ 12:15		2
		12:15 ~ 14:15		3
		14:15 ~ 16:15		4
		16:30 ~ 08:10(03/01/2024)		5
4	Sanding Room	(03/01/2024) 08:38 ~ 10:38	24 hours	1
		10:38 ~ 12:38		2
		12:38 ~ 14:38		3
		14:38 ~ 16:38		4
		16:38 ~ 08:15 (04/01/2024)		5
5	Sanding Room	(04/01/2024) 08:10 ~ 10:10	24 hours	1
		10:10 ~ 12:10		2
		12:10 ~ 14:10		3
		14:10 ~ 16:10		4
		16:10 ~ 08:10 (05/01/2024)		5
5	Air Filter Production	(05/01/2024) 08:20 ~ 10:20	72 hours	1
		10:20 ~ 12:20		2
		12:20 ~ 14:20		3
		14:20 ~ 16:20		4
		16:20 ~ 08:12 (08/01/2024)		5
6	Air Filter Production	(08/01/2024) 08:15 ~ 10:15	24 hours	1
		10:15 ~ 12:15		2
		12:15 ~ 14:15		3
		14:15 ~ 16:15		4
		16:15 ~ 08:15 (09/01/2024)		5
7	Air Filter Production (Only Round HEPA H13)	(09/01/2024) 08:15 ~ 10:15	24 hours	1
		10:15 ~ 12:15		2
		12:15 ~ 14:15		3
		14:15 ~ 16:15		4
		16:15 ~ 08:10 (10/01/2024)		5
8	Air Filter Production (Only Round HEPA H13)	(10/01/2024) 08:10 ~ 10:10	7 hours	1
		10:10 ~ 12:10		2
		12:10 ~ 14:10		3
		14:10 ~ 15:10 (10/01/2024)		4

Figure 15: Air Cleaner Test for Experiment 1

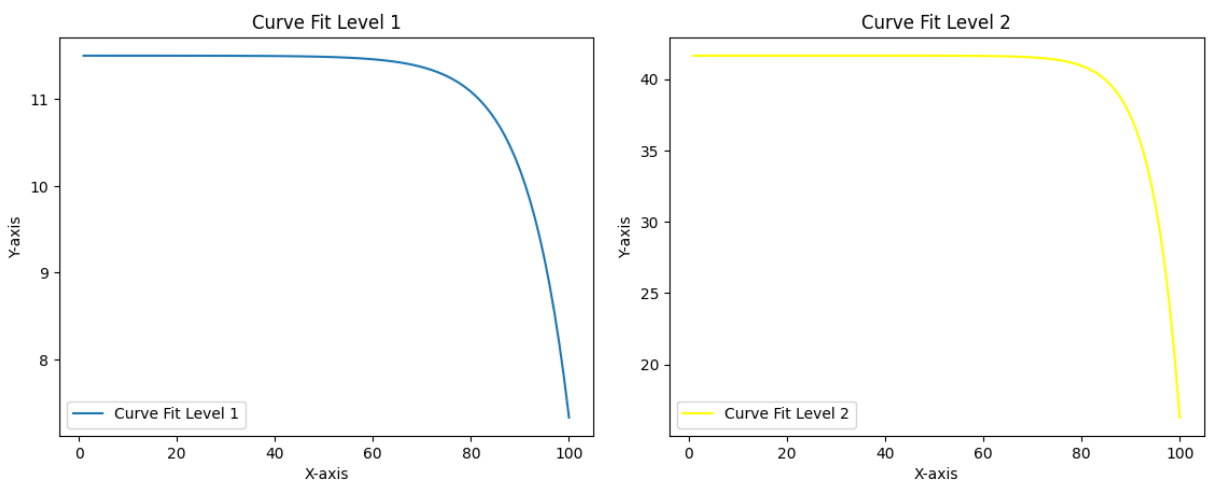
Model: M001 (filter 88cm)	Levels	Fan Speed (RPM)	Fan Module Temperature [°C]:	Fan Motor Temperature [°C]:	Fan Electronics Temperature [°C]:	Temperature	Column1
Normal including Pre-Filter	1	616	36	169	43	28	08/01/2024 13:20
	2	975	37	169	41	27	08/01/2024 13:25
	3	1335	38	169	41	26	08/01/2024 13:30
	4	1695	41	169	41	26	08/01/2024 13:35
	5	2054	47	169	41	26	08/01/2024 13:40
	1	616	42	169	44	27	08/01/2024 13:45
Blocked 25% (22cm)	1	616	33	169	37	24	08/01/2024 16:35
	2	975	35	169	37	24	08/01/2024 16:40
	3	1335	37	169	37	23	08/01/2024 16:45
	4	1685	39	169	38	23	08/01/2024 16:50
	5	2054	40	169	38	23	08/01/2024 16:55
	1	616	39	169	40	23	08/01/2024 17:00
Blocked 50% (44cm)	1	616	35	169	38	23	09/01/2024 07:45
	2	975	36	169	38	23	09/01/2024 08:02
	3	1335	38	169	38	23	09/01/2024 08:13
	4	1695	40	169	39	23	09/01/2024 08:20
	5	2054	44	169	40	23	09/01/2024 08:26
	1	616	40	169	42	24	09/01/2024 08:32
Blocked 75% (66cm)	1	616	35	169	39	24	09/01/2024 09:00
	2	975	36	169	39	24	09/01/2024 09:07
	3	1335	38	169	39	24	09/01/2024 09:15
	4	1695	40	169	39	24	09/01/2024 09:20
	5	2054	45	168	41	24	09/01/2024 09:27
	1	616	40	169	42	24	09/01/2024 09:32
Blocked 100% (88cm)	1	616	37	169	42	25	09/01/2024 09:43
	2	975	36	169	41	24	09/01/2024 09:49
	3	1335	36	169	40	25	09/01/2024 09:55
	4	1695	36	169	39	25	09/01/2024 10:00
	5	2054	37	169	39	25	09/01/2024 10:06
	1	616	37	169	39	25	09/01/2024 10:11

Figure 16: Air Cleaner Test Results for Experiment 2a

(84cm)	1	616	35	169	39	25	11/01/2024 07:09
	2	975	36	169	39	25	11/01/2024 07:15
	3	1335	36	169	38	25	11/01/2024 07:20
	4	2695	37	169	39	25	11/01/2024 07:26
	5	2054	42	169	39	25	11/01/2024 07:31
	1	616	39	169	41	25	11/01/2024 07:36
(79cm)	1	616	38	169	41	25	11/01/2024 07:44
	2	975	38	169	41	25	11/01/2024 07:55
	3	1335	39	169	40	25	11/01/2024 08:03
	4	2054	41	169	41	24	11/01/2024 08:11
	5	2695	42	169	44	24	11/01/2024 08:24
	1	616	41	169	43	24	11/01/2024 08:10
71(cm)	1	616	39	169	43	25	11/01/2024 08:47
	2	975	39	169	42	25	11/01/2024 08:53
	3	1335	40	169	41	25	11/01/2024 09:04
	4	2054	41	169	41	24	11/01/2024 09:11
	5	2695	45	169	42	24	11/01/2024 09:16
	1	616	40	169	43	24	11/01/2024 09:20
59(cm)	1	616	38	169	42	24	11/01/2024 09:26
	2	975	38	169	41	24	09:50
	3	1335	40	169	31	24	09:56
	4	2054	41	169	41	24	10:03
	5	2695	45	169	42	24	10:08
	1	616	43	169	43	24	10:15
51(cm)	1	616	41	169	44	24	10:17
	2	975	41	169	43	24	10:23
	3	1335	42	169	43	24	10:29
	4	2054	42	169	43	24	10:35
	5	2695	47	169	44	24	10:40
	1	616		169			10.45

Figure 17: Air Cleaner Test Results for Experiment 2b

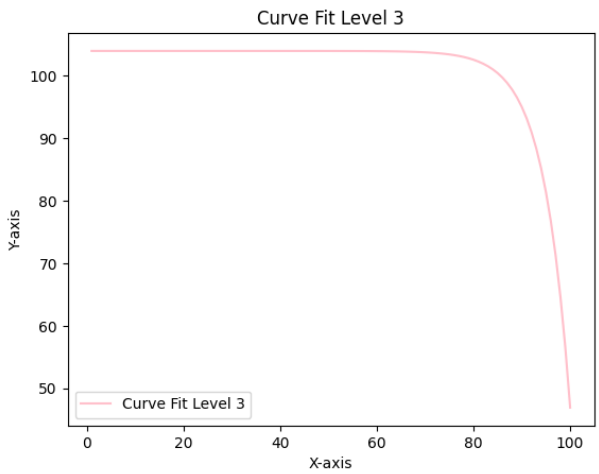
7.3 Curve-Fit Functions for each level



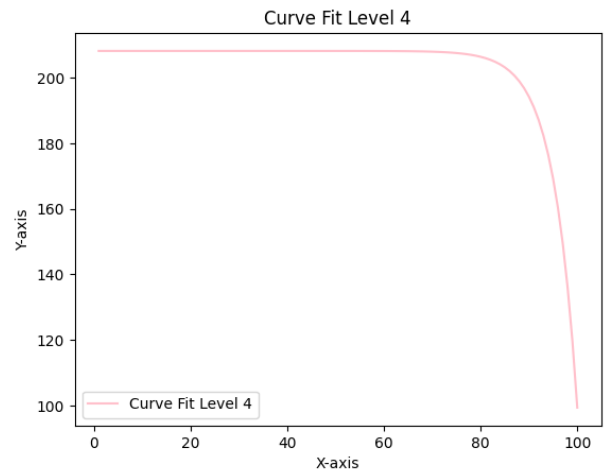
(a) Curve-fit function for operational level 1

(b) Curve-fit function for operational level 2

Figure 18: Snippet of curve-fit function for operational 1 and 2



(a) Curve-fit function for operational level 3



(b) Curve-fit function for operational level 4

Figure 19: Snippet of curve-fit function for operational 3 and 4

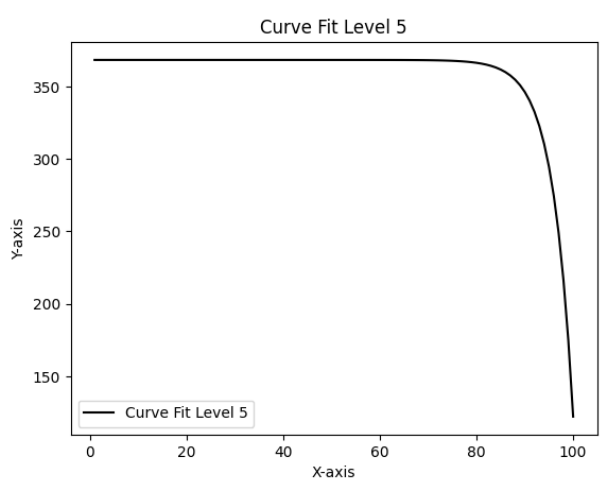


Figure 20: Curve-fit function for operational level 5

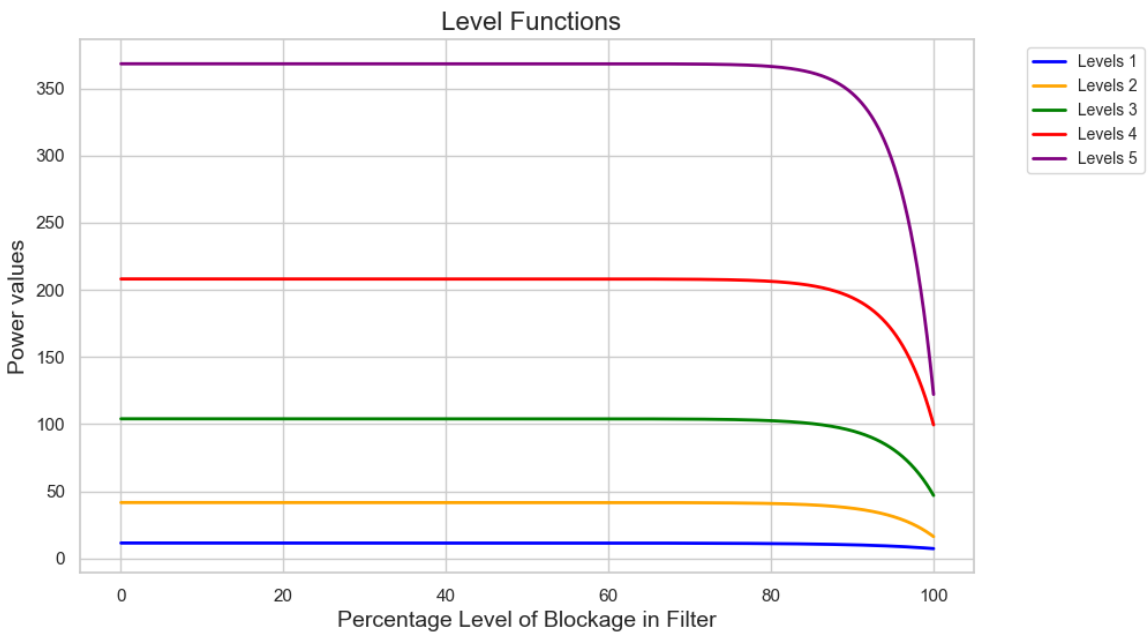


Figure 21: Combined graph of all curve-fit functions

Sensor Data

PM 1.0 [µg/m³]:	1
PM 2.5 [µg/m³]:	4
PM 10 [µg/m³]:	6
TVOC [ppb]:	162
eCO_2 [ppm]:	643
NO_2 [ppb]:	0
Temperature [°C]:	27
Relative Humidity [%]:	71
Fan Power [W]:	11
Fan Speed [RPM]:	616 / 616

Fan Data

Fan Model:	M001
Total Fan Hours On [Hours]:	3519
Fan Maximum Speed [RPM]:	2630
Fan Module Temperature [°C]:	39
Fan Motor Temperature [°C]:	169
Fan Electronics Temperature [°C]:	43

Firmware

Version:	0.925 (2023-11-21)
AC Id:	EB2A3A8

Figure 22: localhost data

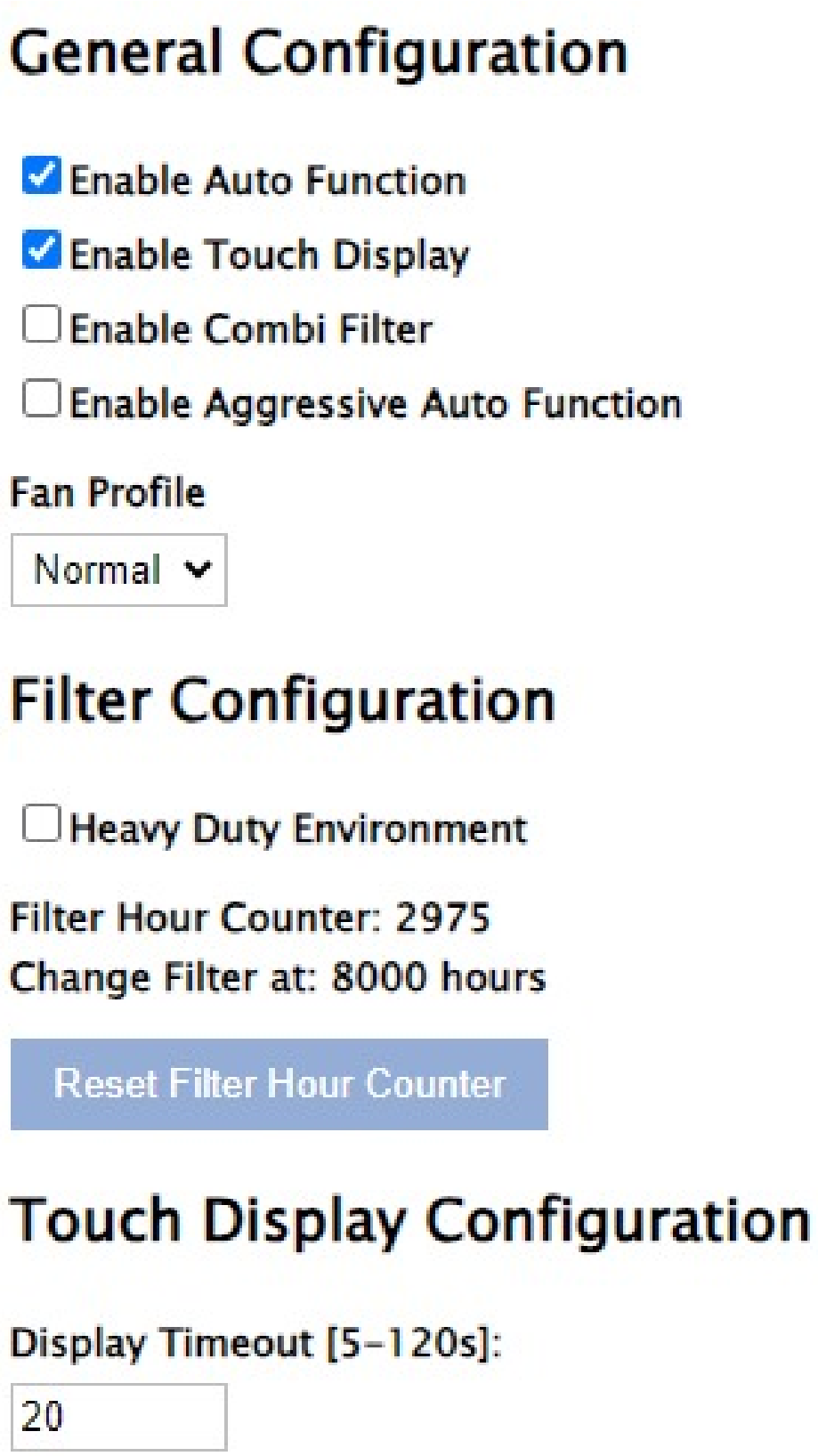


Figure 23: localhost settings options

Figure 24: localhost WIFI Configuration